

HOSPITAL MANAGEMENT SYSTEM

USE CASE FLOW

Goal	UC-01 – Patient Registration & Login
Actor	Patient
Main flow	1. Patient opens the system. 2. Patient selects Register. 3. Patient enters name, age, sex, phone number and email. 4. System validates the entered details. 5. System sends an OTP to the registered phone number. 6. Patient enters the OTP. 7. System verifies the OTP and creates the patient account. 8. Patient enters valid login credentials. 9. System authenticates the patient and provides access.
Alternate Flow	At Step 4, if the entered details are invalid: 4a. System displays an error message. 4b. Patient corrects and re-enters the invalid details. 4c. System returns to Step 4 to validate the corrected details.

Goal	UC-02 – OTP Verification
Actor	Patient
Main flow	1. Patient provides the registered phone number. 2. System generates and sends an OTP. 3. Patient enters the received OTP. 4. System validates the OTP. 5. If the OTP is correct, the phone number is verified. 6. System allows the patient to continue registration/login.
Alternate Flow	At Step 4, if the OTP is incorrect: 4a. System displays an error message. 4b. Patient re-enters the OTP. 4c. System returns to Step 4 to validate the new OTP.

Goal	UC-03 – View Medical History
Actor	Patient / Doctor
Main flow	1. User logs into the system. 2. User selects the patient's medical history. 3. System verifies user authorization. 4. System retrieves the patient's stored medical history and reports. 5. System displays the available medical information.
Alternate Flow	At Step 3, if the user is not authorized: 3a. System displays an access-denied message. 3b. User cannot access the patient's medical information. 3c. Use case ends.

Goal	UC-04 – Doctor Availability & Appointment Booking
Actor	Patient
Main flow	1. Patient logs into the system. 2. Patient selects the required medical field. 3. System displays available doctors. 4. Patient selects a doctor. 5. System displays available appointment schedules.

	6. Patient selects an available slot. 7. System confirms and stores the appointment. 8. System displays the appointment confirmation.
Alternate Flow	At Step 5, if no appointment slot is available: 5a. System informs the patient that no slot is available. 5b. System displays other available schedules/doctors. 5c. Patient selects another doctor or schedule and returns to Step 4.

Goal	UC-05 – Appointment Cancellation
Actor	Patient
Main flow	1. Patient logs into the system. 2. Patient views booked appointments. 3. Patient selects an appointment to cancel. 4. System displays the appointment details. 5. Patient confirms cancellation. 6. System updates the appointment status to cancelled. 7. System makes the slot available again.
Alternate Flow	At Step 5, if the patient does not confirm cancellation: 5a. System keeps the appointment active. 5b. System returns the patient to the booked appointments at Step 2. 5c. Patient may select another appointment or end the process.

Goal	UC-06 – Medical Diagnosis & Reports
Actor	Doctor
Main flow	1. Doctor logs into the system. 2. Doctor selects the patient's appointment/record. 3. System displays the patient's available medical history. 4. Doctor examines the patient. 5. Doctor enters the initial diagnosis. 6. Doctor specifies the required reports/scans. 7. System saves the diagnosis and required reports/scans to the patient's record.
Alternate Flow	At Step 3, if required patient information is unavailable: 3a. System displays the available information. 3b. Doctor proceeds with the available information and enters the diagnosis. 3c. Flow continues to Step 5.

Goal	UC-07 – Prescription Management
Actor	Doctor / Patient
Main flow	1. Doctor logs into the system. 2. Doctor opens the patient's medical record. 3. Doctor enters the final prescription and medication details. 4. System validates and stores the prescription. 5. Patient logs into the system. 6. Patient opens the prescription section. 7. System displays the stored prescription.
Alternate Flow	At Step 4, if prescription details are incomplete: 4a. System identifies the incomplete prescription details. 4b. System asks the doctor to provide the required information. 4c. Doctor corrects the prescription and returns to Step 3.

Goal	UC-08 – Online Payment & Medicine Purchase
Actor	Patient
Main flow	1. Patient views the prescribed medicines. 2. Patient selects the required medicines. 3. System calculates the payment amount. 4. Patient selects online payment. 5. Patient provides the required payment details. 6. System processes the payment securely. 7. System confirms successful payment. 8. System records the medicine purchase.
Alternate Flow	At Step 6, if payment fails: 6a. System displays the payment failure message. 6b. Patient retries the payment using the payment details at Step 5. 6c. System returns to Step 6 to process the payment again.

Goal	UC-09 – File Download
Actor	Patient
Main flow	1. Patient logs into the system. 2. Patient opens the medical documents section. 3. System displays available reports, prescriptions and other files. 4. Patient selects the required file. 5. Patient selects Download. 6. System downloads the selected file to the patient's device.
Alternate Flow	At Step 5, if the selected file is unavailable: 5a. System displays an appropriate error message. 5b. Patient selects another available file at Step 4. 5c. System returns to Step 5 to download the selected file.

Goal	UC-10 – Check Bed Availability
Actor	Hospital Staff / Patient
Main flow	1. User accesses the bed availability section. 2. System retrieves the current bed information. 3. System checks the status of available beds. 4. System displays the current bed availability as available or occupied.
Alternate Flow	At Step 3, if no beds are available: 3a. System displays that all required beds are currently occupied. 3b. User returns to Step 1 to check bed availability again later. 3c. Use case ends if no further check is required.